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Faculty of
Computing

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ARTIFICIAL INTELLIGENCE (SECJ3553)

SECTION 03

GROUP 2

ASSIGNMENT 1:

EcoRoute

THEME:

SMART CITY

LECTURER:

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Knowledge Representation

1. **High-Priority Bin Collection:** Keep monitoring the bin until it exceeds a certain level of amount.

*IF (bin_fill_level("Bin_123") > 85%) AND (bin_status("Bin_123") != "Damaged")
THEN add_to_route("Bin_123", "HighPriority")*

Explanation:

This KR is solving the City Council's primary need for operational efficiency. EcoRoute uses its sensors to continuously check each bin's fill level. According to this rule, a bin should only be taken into consideration for a collection route if its fill level exceeds a high threshold of 85%. The outdated, ineffective method of maintaining empty or barely used bins is immediately stopped by this logic. The AI can determine the most effective way to service only these high-value targets by generating "HighPriority" tasks exclusively for full bins. Another level of intelligence is added by the AND condition ($\neq$ "Damaged"), which stops the system from wasting a truck's time by sending it to an unusable location. This prevents unsuccessful collection attempts, which are a significant reason for fuel and workers wasted.

Predicate Logic/ First Order Logic:

$$\forall x((FillLevel(x) > 0.85) \wedge \neg bin_status(x, "Damaged")) \\ \rightarrow AddRoute(x, HighPriority)$$

For any bin (x), if the fill level of bin (b) is greater than 85, AND the status of bin (b) is NOT "Damaged", THEN the system will add bin (b) to the collection route with "HighPriority".

2. **Predictive Overflow Prevention:** Proactively monitoring bins during peak periods like weekends and public holidays.

*IF (current_day = "Weekend" OR "PublicHoliday") AND
(historical_overflow_risk("Bin_123", current_day) = "High") AND
(bin_fill_level("Bin_123") > 60%) THEN add_to_route("Bin_123",
"MediumPriority")*

This KR focuses on preventing bin overflow by leveraging predictive insights and contextual factors such as weekends and public holidays. The technology analyzes past data and current sensor readings to identify bins that are likely to overflow during high-demand days. According to this rule, if the current day is a weekend or public holiday, and the historical overflow risk for a bin like “Bin_123” is recorded as “High,” and its current fill level exceeds 60%, then the system will automatically add the bin to the collection route with “MediumPriority.” By ensuring that trash cans in crowded places are serviced before they overflow, this preventive thinking improves cleanliness and reduces citizen complaints.

Explanation:

This rule directly solves the resident's need. Instead of waiting for the bin to hit 85%, this rule uses three conditions:

1. Calendar: It checks if today is a "Weekend" or "PublicHoliday".
2. Historical Data: It checks its memory to see if this specific bin has a "High" risk of overflowing on these days.
3. Lower Threshold: It uses a lower fill level (> 60%) to act before the bin becomes critical.

Predicate Logic/ First Order Logic (FOL):

$$\begin{aligned} \forall x \forall d (\text{OnDay}(d) \wedge (\text{Weekend}(d) \vee \text{PublicHoliday}(d)) \\ \wedge \text{HistoricalOverflowRisk}(x, \text{"High"}) \wedge \text{FillLevel}(x) > 0.60) \\ \rightarrow \text{AddRoute}(x, \text{"MediumPriority"}) \end{aligned}$$

For any day (d) and any bin (x), if the day (d) is a Weekend OR a Public Holiday, AND the historical overflow risk for that bin (x) is "High", AND the fill level of bin (x) is greater than 0.6 (60%), THEN the system will add bin (x) to the collection route with "MediumPriority". This enables predictive scheduling that helps the waste management system handle peak times efficiently and avoid overflow incidents before they occur.

3. **Commercial Zone Priority:** If a bin is in a commercial zone and fill level >70%, assign it to the nearest available truck.

*IF Location("Bin_123", "Commercial") AND FillLevel("Bin_123") > 0.70
THEN AssignTruck("Bin_123", NearestTruck)*

Explanation:

EcoRoute uses GPS to classify all bins by zone type which are residential, commercial or industry. Commercial areas like City Square or restaurants generate significantly more waste daily than homes. This rule triggers when a commercial bin fill level hits 70%, a threshold lower than residential to account for faster filling. For example, Bin_789 near a food court at 75% full, then system automatically locates the nearest available truck. This minimizes response time, prevents overflow in high-traffic public areas and balances truck workload. It directly supports the city council's goal of efficient fleet management.

Predicate Logic / First Order Logic (FOL):

$$\forall x (Location(x, "Commercial") \wedge FillLevel(x) > 0.70) \\ \rightarrow AssignTruck(x, NearestAvailable)$$

For any bin (x), if it is located at a commercial zone AND its fill level exceeds 70%, then assign it to the nearest available truck. The $\forall x$ applies to all commercial bins. The $\wedge$ ensures both location and fill conditions. The $\rightarrow$ triggers dynamic truck assignment. This support real-time efficiency and driver workload balance.

4. **Driver Route Notification:** If a route is generated and the truck is idle, send the route to the driver's tablet.

*IF RouteGenerated(Truck_A) AND TruckStatus(Truck_A, "Idle")
THEN SendRoute(Truck_A, DriverApp)*

Explanation:

EcoRoute generates optimized routes only when needed and sends them instantly via mobile app to drivers. This rule will wait for the truck to complete its current task and enter "Idle" status before pushing the new route. Then, driver receives a clear, interactive map: "Bin_123 to Bin_456 to Depot (45 min, 12 km)" with turn-by-turn GPS. By doing so, no more paper schedules, radio calls, or confusion. This improves driver satisfaction, reduces downtime and ensures real-time execution of dynamic plans. It supports the driver's need for shorter and smarter daily routes.

Predicate Logic / First Order Logic (FOL):

$$\forall t (RouteGenerated(t) \wedge TruckStatus(t, "Idle")) \\ \rightarrow SendRoute(t, DriverApp)$$

For any truck (t), if a route has been generated for (t) AND (t) is currently idle, then send the route to the driver's app. The $\forall t$ applies to all 20 trucks in the fleet. The $\wedge$ ensures timing is correct. The $\rightarrow$ triggers push notification. This completes the on-demand collection loop.

- 5. Truck Overloading Prevention:** Detect current truck capacity to avoid from assigning it to another bin when maximum capacity is nearly reached.

IF (TruckLoad ("Truck_A") > 90% THEN ReassignRemainingBins ("Truck_A", NearestAvailableTruck)

Explanation:

EcoRoute uses a weight sensor to continuously check current load percentage of the waste collection truck. This rule is used as an indicator to reassign the bin to be collected, when the current assigned truck's load surpasses the threshold of 90%. It will find the nearest available truck and transfer all remaining bins of Truck_A as Truck_A will immediately stop collecting bins. The system ensures a safe waste transportation and increases efficiency by preventing minimizing unexpected delays or hazard, such as mechanical strain and braking problems due to overweight truck.

Predicate Logic/ First Order Logic (FOL):

$\forall t (TruckLoad(t) > 0.90) \rightarrow ReassignRoute(t, NearestAvailableTruck)$

For any truck (t), IF the truckload is over 0.90, THEN the system will reassign the bin route from truck (t) to the nearest truck which is available. This ensures a safe operating limit and prevent overloading incidents.

6. **Route Optimization Trigger:** If more than 5 high-priority bins are located in the same zone, automatically generate an optimized collection route using the Traveling Salesperson Problem (TSP) algorithm.

*IF CountHighPriorityBins (Zone_A) > 5
THEN GenerateRoute (Truck_X, TSP_Algorithm)*

Explanation:

EcoRoute monitors bin clusters every hour using GPS coordinates. When 6 or more high-priority bins appear in one zone, for example, Taman Ungku Tun Aminah night market area, the fixed-route system becomes highly inefficient. This rule triggers the TSP algorithm to calculate the shortest possible path that visits all full bins and returns to the depot. For example, 8 bins in Zone_A that used to take 21 km under fixed schedule are reduced to only 13 km. This single rule can save 2 to 3 hours and 30 to 40% fuel per truck per day, directly contributing to the city council's budget reduction goal.

Predicate Logic/ First Order Logic (FOL):

$$\forall z \forall t (Zone(z) \wedge CountHighPriority(z) > 5 \wedge AvailableTruck(t)) \\ \rightarrow GenerateRoute(t, z, TSP)$$

For any zone (z) and truck (t), if the zone contains more than 5 high-priority bins AND truck (t) is available, then generate an optimized route using TSP. The quantifiers $\forall z \forall t$ cover all zones and trucks and $\wedge$ combines the conditions. The $\rightarrow$ activates the optimization engine. This rule transforms scattered collection points into the shortest possible path.

Knowledge Representation Involved to Achieve Goals

EcoRoute uses a declarative, hybrid knowledge representation strategy focused on providing actionable, real-time waste management decisions. This is achieved primarily through Logical Representation, specifically First-Order Logic, FOL and Production Rules. This combination forms the core of EcoRoute, enabling it to do more than just monitor bins and instead make smart decisions for waste collection routes.

Logical Representation (FOL)

EcoRoute employs First-Order Logic to formally describe relationships, constraints, and conditions between bins, trucks, zones, and times. For example, it can express rules like:

$$\forall x((\text{FillLevel}(x) > 0.85) \wedge \neg \text{bin_status}(x, \text{"Damaged"})) \\ \rightarrow \text{AddRoute}(x, \text{HighPriority}))$$

This ensures that the system is logically consistent, clearly defines its domain, and allows precise reasoning about operational decisions. It provides a solid foundation to model complex relationships, such as predicting overflow, prioritising bins in commercial zones, and assigning trucks optimally.

Production Rules

Production Rules serve as the operational mechanism to execute actions based on conditions. These are implemented as IF-THEN rules, defining thresholds and triggers for tasks such as High-Priority Bin Collection, Predictive Overflow Prevention, and Driver Route Notification.

For example:

*IF Location("Bin_123", "Commercial") AND FillLevel("Bin_123") > 0.70
THEN AssignTruck("Bin_123", NearestTruck)*

This approach is modular, interpretable, and easy to maintain, allowing new rules to be added as city requirements change. Production Rules, combined with FOL, enable EcoRoute to act intelligently in real time, dynamically optimizing collection routes and reducing fuel consumption, operational costs, and overflow incidents.

Assignment 1 Peer Review Form

Evaluator: Choh Jing Yi

Name	Rating (1-4)	Notes/Comments (if any)
Brendan Chia Yan Fei	4	
Anis Safiyya Binti Janai	4	
Lim Yu Han	4	

Evaluator: Brendan Chia Yan Fei

Name	Rating (1-4)	Notes/Comments (if any)
Choh Jing Yi	4	
Anis Safiyya Binti Janai	4	
Lim Yu Han	4	

Evaluator: Anis Safiyya Binti Janai

Name	Rating (1-4)	Notes/Comments (if any)
Choh Jing Yi	4	
Brendan Chia Yan Fei	4	
Lim Yu Han	4	

Evaluator: Lim Yu Han

Name	Rating (1-4)	Notes/Comments (if any)
Choh Jing Yi	4	
Brendan Chia Yan Fei	4	
Anis Safiyya Binti Janai	3	